A Festive 3D Greeting Card!

Computer Graphics Project

F14

Done by

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Introduction

The project aims to create an interactive 3D greeting card scene that captures the joy of celebration. At the center is a table with a three-layer cake, surrounded by gifts, party decorations, and balloons. The scene also features congratulatory phrases and colorful firecrackers that enhance the festive mood. Users can control the lighting and trigger animations, making the experience lively and engaging.

The objectives include implementing 3D shapes, texture mapping, dynamic lighting, and interactive features that enhance user engagement with the scene.

Implementation Details

3D Shapes and Texture Mapping

We used Cylinder for the cake and table, and Box for the gifts and the carpet, applying Phong materials for realistic textures. Each shape was positioned thoughtfully to create a consistent design.

Lighting Sources

We implemented light to illuminate the scene. Lighting was controlled via user input, allowing the user to toggle lights on/off with the 'O' and 'F' keys.

Implementation Details

User Interactivity

Key event were added to capture user inputs:

- Pressing 'O' turns on the lights, while 'F' turns them off.
- Pressing the 'Mouse click' key triggers an animation displaying the phrase "Congratulations" accompanied by colorful firecracker effects.

3D Transformations

Various transformations were applied to the 3D shapes to create a more dynamic scene.

Translation on X, Y, and Z.

challenges faced

A major challenge was getting each element in the right spot. This meant making sure the cake was centered on the table, the gifts looked nice around it, and the decorations were placed well. It required careful adjustments to their positions.

solutions implemented: we used Figma to create a layout design for the scene. This allowed for precise positioning and visual planning of where each object (cake, gifts, decorations) would be placed in the 3D environment.

User Guide

Moving Objects:

X-axis: Move objects left or right.

Y-axis: Move objects up or down.

Z-axis: Move objects closer or further away from the viewer.

Lighting Control:

Press 'O': Turns on the lights in the scene.

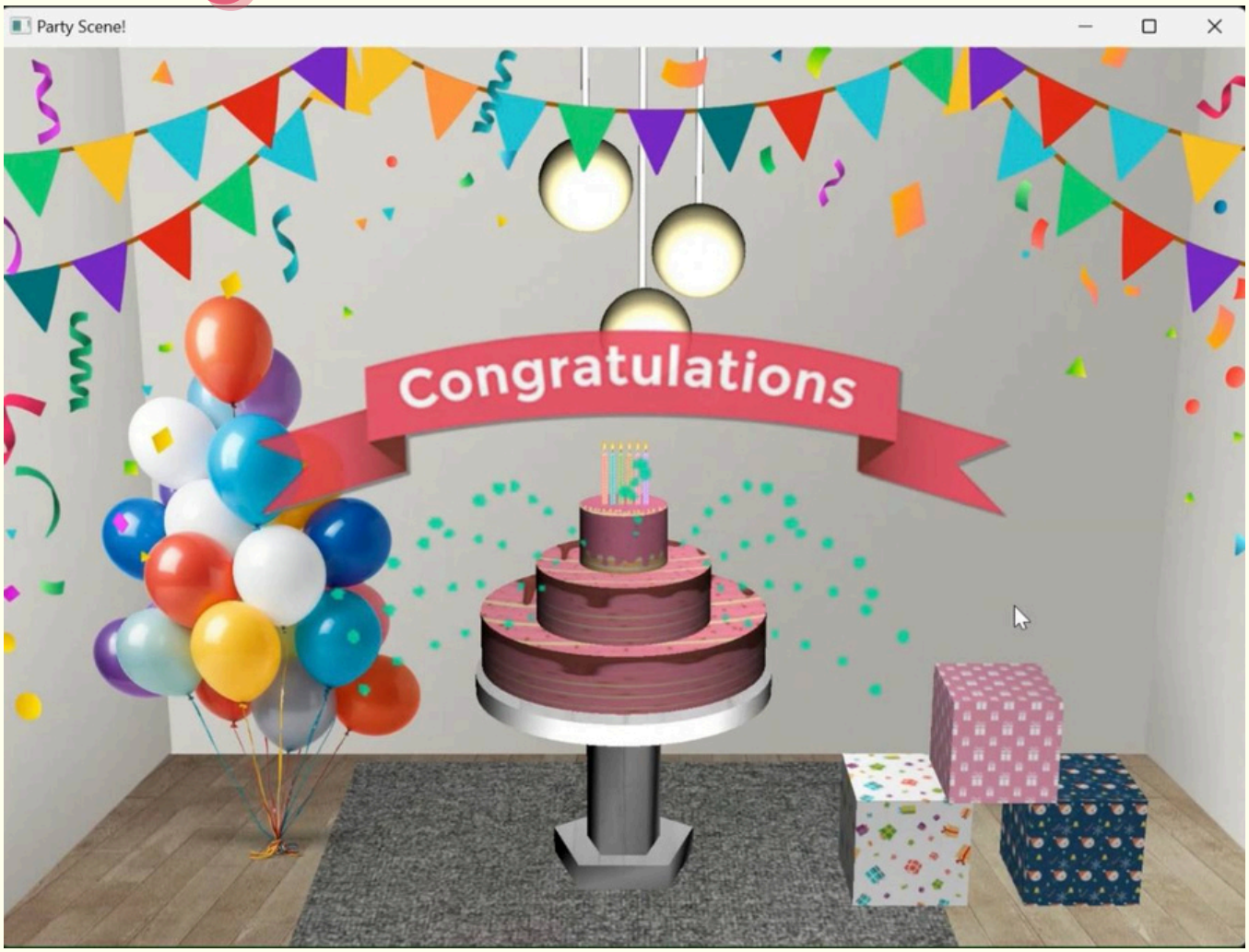
Press 'F': Turns off all lights.

Celebration Activation:

Click 'Mouse': Displays the message "Congratulations" with colorful firecracker animations.

Project Demo

Project screenshots

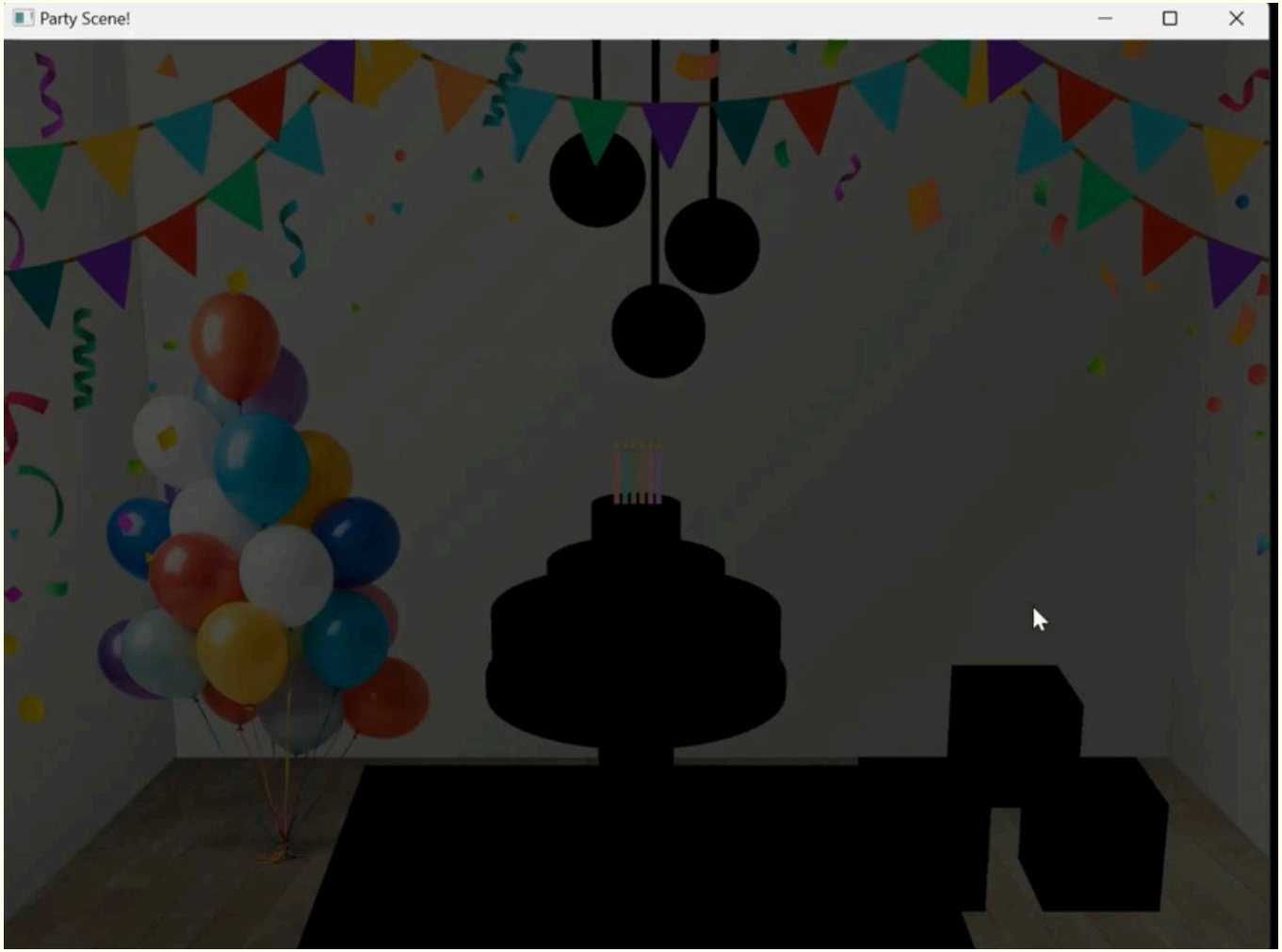


When the user 'clicks the mouse' a burst of colorful firecrackers will light up the screen, accompanied by joyful congratulatory messages!

Project screenshots

When the user presses 'O', the lights will turn on, brightening the scene and adding to the festive mood!

Project screenshots



When the user presses 'F', the lights will turn off, darkening the scene.

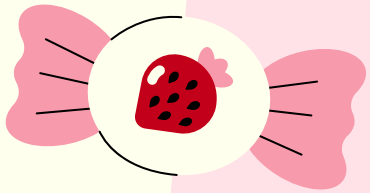
Future Enhancements

Enhanced Visuals

Incorporating more intricate decorations and animations to enhance the scene.

Sound Effects

Adding background music and sound effects for interactions to increase enjoyable.



Conclusion

This project provided valuable insights into 3D graphics programming and user interaction design. We learned how to effectively use JavaFX for creating immersive environments and how to manage user inputs for interactivity.

Resources

- Figma: Used for designing the layout and positioning of elements in the scene.
- Freepik: Provided texture images for enhancing the realism of 3D objects.
- Pinterest: Sourced background images and inspiration for decorations and overall scene design.

Thank you for listening